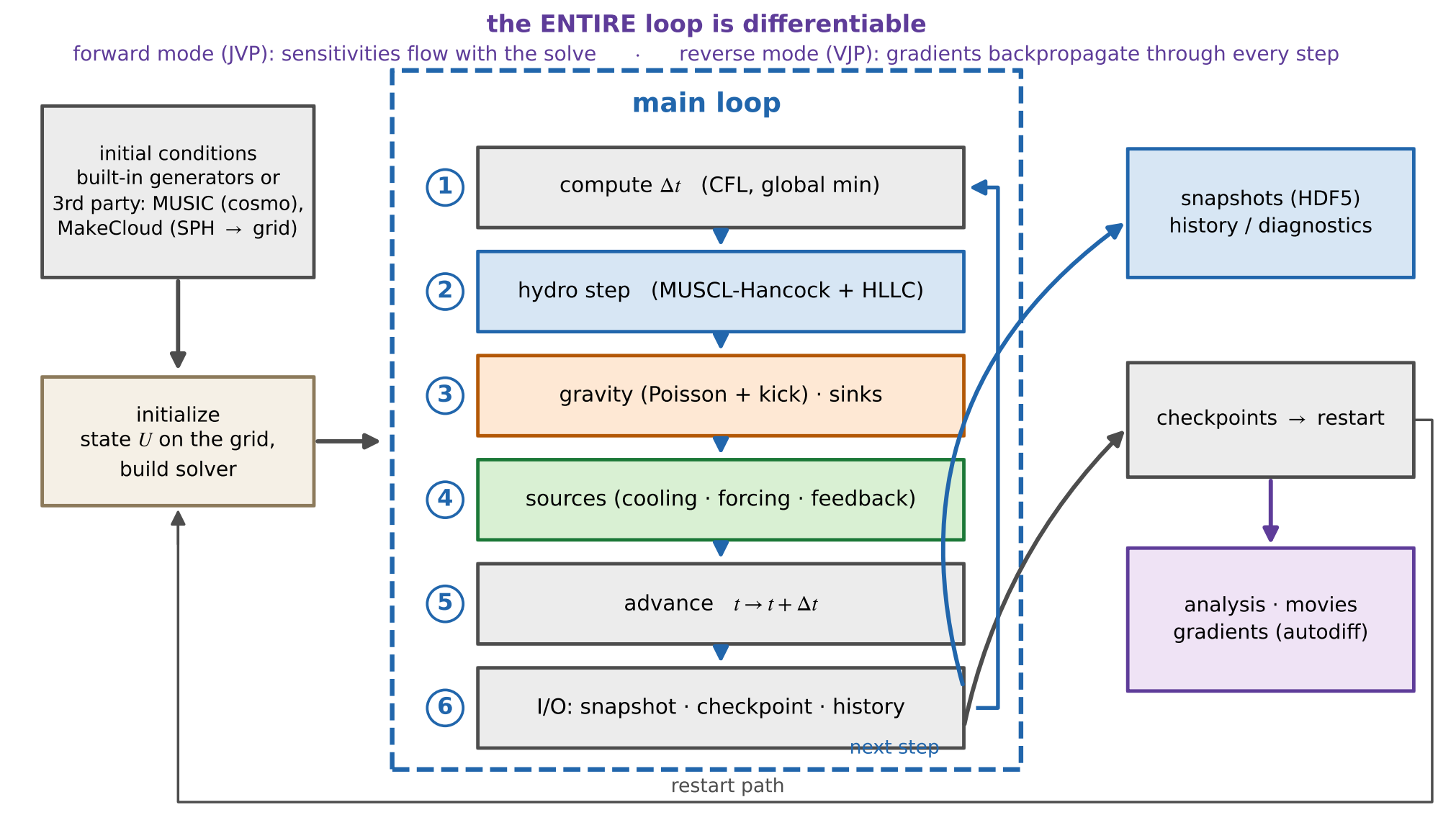
the ENTIRE loop is differentiable
forward mode (JVP): sensitivities flow with the solve · reverse mode (VJP): gradients backpropagate through every step
main loop
initial conditions
built-in generators or
3rd party: MUSIC (cosmo),
MakeCloud (SPH → grid)
initialize
state U on the grid,
build solver
1
compute Δt (CFL, global min)
2
hydro step (MUSCL-Hancock + HLLC)
3
gravity (Poisson + kick) · sinks
4
sources (cooling · forcing · feedback)
5
advance $t \to t + \Delta t$
6
I/O: snapshot · checkpoint · history
next step
snapshots (HDF5)
history / diagnostics
checkpoints → restart
analysis · movies
gradients (autodiff)
restart path